For Learning, Toil is Not Optional

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Recently while browsing the internet, I was served an ad by Anthropic—the creators of Claude—which claimed that "toil is optional." Ordinarily I ignore sensationalist PR slop, but as a software developer I recognise that it is not entirely hyperbole. As of late 2025, coding agents such as Claude Code and Github Copilot demonstrate capabilities that would have been difficult to fathom even a few years ago.

> **CLAUDE**
> Claude is a large language model developed by Anthropic, a US AI safety company founded in 2021. Anthropic positions Claude as a "Constitutional AI" system — one trained with explicit guidelines intended to make its outputs more honest and less harmful than models trained purely on human feedback. Claude Code is its agentic variant, capable of reading, writing, and executing code autonomously within a developer's environment.

> **CODING AGENTS**
> A coding agent is an AI system capable of autonomously completing multi-step software development tasks: writing functions, running tests, reading error messages, and iterating toward a working solution — all without continuous human direction. Unlike a simple autocomplete tool, an agent can plan, act, observe the result, and revise. Claude Code and GitHub Copilot (in its agentic modes) are leading examples as of 2025.

Much of the routine burden of software development—repeating familiar patterns, tracking down elusive bugs, maintaining documentation, and managing dependencies—is being reduced at a rapid pace.

> **REPEATING FAMILIAR PATTERNS**
> In software development, a *design pattern* is a reusable solution to a commonly occurring problem — things like "how do I connect to a database?" or "how do I handle user authentication?" Experienced developers recognise these situations instantly and apply standard templates. For an LLM trained on millions of codebases, reproducing these patterns is trivially easy. The cognitive work a junior developer would do learning the pattern is simply bypassed.

> **MAINTAINING DOCUMENTATION**
> Documentation is the written record that explains what code does, why it was written that way, and how other developers should use it. Keeping it accurate as code changes is tedious and frequently neglected. LLMs can read a function and generate a plausible docstring in seconds — one of the clearest cases where AI reduces genuine toil without obvious learning cost, because writing docs rarely builds new understanding.

> **MANAGING DEPENDENCIES**
> Modern software projects rely on dozens or hundreds of external libraries — packages written by others that your code imports and uses. Managing these involves tracking versions, resolving conflicts when two libraries require incompatible versions of a third, and applying security patches. This process (sometimes called "dependency hell") is mechanical and time-consuming. Tools like npm, pip, and cargo automate much of it; AI assistants can now diagnose and resolve conflicts that would previously require significant developer effort.

While toil in software development is increasingly viewed as a liability to be minimised, with education it's a different story because toil plays a completely different role. Far from being a liability, it is formative. It is indispensable. The effort involved in grappling with material is inseperable from the biological processes through which understanding is built. When that struggle is offloaded, we remove the conditions under which learning can occur and mastery can develop.

A 2025 study from the MIT Media Lab on AI-assisted writing found clear differences in brain activity[1]:

> *EEG revealed significant differences in brain connectivity: Brain-only participants exhibited the strongest, most distributed networks; search engine users showed moderate engagement; and LLM users displayed the weakest connectivity.*

Participants wrote an essay either unaided, with search engines, or with LLMs. While the study measures electrical activity via Electroencephalography, prior developmental research suggests that changes in EEG connectivity can track (though not directly measure) underlying processes such as Synaptogenesis and Myelination[2]. Any observed reduction in neural engagement should therefore be interpreted cautiously: it may indicate less coordinated or less demanding network activity, which could correspond to conditions that are less conducive to synaptogenesis, rather than implying a direct reduction in synaptogenesis itself.

> BRAIN ACTIVITY (EEG CONNECTIVITY)
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> The MIT study measured *functional connectivity* — how strongly different brain regions activate together, rather than in isolation. Higher connectivity suggests the brain is coordinating more widely distributed networks to complete a task. When LLM users showed the weakest connectivity, it means their brains were engaging fewer regions in a less coordinated way — consistent with less effortful, less generative processing.

Electroencephalography (EEG)

Synaptogenesis

Synaptogenesis: New Synapse Formation

Myelination

Myelination: Myelin Sheath on Axon

While this relationship is not definitively established, for the purposes of a blog-level argument, the connection is sufficiently well-supported to justify a cautious proxy: I am comfortable treating reduced EEG engagement as indicative of reduced conditions for synaptic development, given the broader claim that learning is metabolically and cognitively effortful—that it requires a degree of toil to drive the neural changes we associate with durable understanding. See Bjork & Bjork (2020)[3] for more on that.

With that disclaimer established, now imagine a field of tall grass. Each time you walk between two landmarks—a boulder, a stream, a stand of trees—you press down a faint trail. That is synaptogenesis: the path begins to exist. Walk it repeatedly, and the trail hardens, the ground compacts, and movement becomes faster. That is myelination: the path becomes efficient. A brain that labours lays down paths and reinforces them. A brain that offloads too early walks less ground—and builds fewer roads. Mastery requires walking those paths again and again.

SYNAPTOGENESIS — THE PATH BEGINS TO EXIST

In the metaphor: pressing down a faint trail through tall grass for the first time. In the brain: a new synaptic connection forms between two neurons because they fired together during a cognitive act. Hebb's rule captures this: *neurons that fire together wire together.* The trail doesn't exist until you walk it — and walking it is the struggle.

MYELINATION — THE PATH BECOMES EFFICIENT

In the metaphor: the trail hardens and widens with repeated use, making movement faster. In the brain: glial cells wrap the axon in myelin — a fatty insulating sheath — after a pathway has been activated repeatedly. This increases signal transmission speed by up to 100×. Myelination is why a novice's movements feel effortful while an expert's feel automatic: the expert's pathways are heavily insulated. It is the biological correlate of fluency.

Now consider LLMs. During training, a model traverses an immense number of possible paths between conceptual "landmarks," adjusting its internal weights through repeated passes over data. Modern training runs involve on the order of 10^{26} discrete computations—a scale of effort no human could approximate. This process is not magical; it is iterative and exhaustive. Each pass reinforces certain pathways and weakens others, gradually producing a system that can move reliably between ideas. Without that sustained computational labour, the model would have no coherent way to navigate those relationships at all.

LLMS (LARGE LANGUAGE MODELS)

A Large Language Model is a neural network trained to predict the next token (roughly: the next word or word-fragment) in a sequence, given all preceding tokens. At sufficient scale — billions of parameters trained on trillions of words — this simple objective produces systems capable of reasoning, translation, summarisation, and code generation. The "large" refers both to the number of parameters in the model and the volume of training data consumed.

INTERNAL WEIGHTS

A neural network's *weights* are numerical values — typically 32-bit or 16-bit floating point numbers — stored at every connection between artificial neurons. A model with 70 billion parameters has 70 billion such numbers. During training, an algorithm called *backpropagation* calculates how much each weight contributed to the model's prediction error and adjusts it slightly in the direction that reduces that error. After billions of such adjustments, the weights encode a compressed statistical representation of the training data.

DISCRETE COMPUTATIONS (10^{26})

A *floating point operation* (FLOP) is a single arithmetic calculation — an addition or multiplication of decimal numbers. Training a frontier LLM involves repeating such operations across billions of parameters, billions of tokens, many times. GPT-4 is estimated at roughly 10^{25} FLOPs; more recent models push toward 10^{26}. To put this in perspective: a human brain performs roughly 10^{15}–10^{16} synaptic operations per second. Even at that rate, it would take the brain several million years of continuous operation to match one training run.

Training

LLM Training Loop: Forward Pass → Loss → Backprop

Seen in this light, the apparent fluency of an LLM is the product of accumulated work rather than inherent capability. The same principle holds for human learners. When we look past the surface, what appears effortless is underwritten by repetition and reinforcement. The difference is that, in machines, the toil is hidden in training runs, and in humans, it is hidden in practice. When a learner is handed the finished artifact—the essay—they are given the outcome without engaging in the process that builds those pathways.

TRAINING RUNS

A *training run* is the complete process of training a model from random initialisation to a converged state — from a network of random numbers to one that can answer questions. Large training runs last weeks or months, consuming thousands of specialised chips (GPUs or TPUs) running continuously. The cost of a frontier training run is measured in tens or hundreds of millions of dollars. The trained model — the artifact handed to users — contains none of this visible effort. It simply responds.

PRACTICE (DELIBERATE PRACTICE)

Deliberate practice, as described by Anders Ericsson, is effortful, focused repetition at the edge of current ability — with immediate feedback and correction. It is distinct from mere repetition (doing the same thing many times) and from play (doing something enjoyable without strain). Ericsson's research across music, chess, sport, and medicine found that expert performance is overwhelmingly predicted by accumulated deliberate practice hours, not by innate talent. The neural correlate is the myelination and synaptic consolidation described earlier in this essay.

Returning to our visual metaphor, the paths are not walked. The trails are not cut. The roads are never consolidated. The learner may hold a finished piece of writing, but they remain in the field, without the means to move from one landmark to the next.

Now, what should we make of all this? We cannot simply ban LLMs. Instead, we need to make cognitive toil desirable again. This runs against the prevailing framing of such tools, where toil is treated as inefficiency to be eliminated. But the evidence suggests otherwise: when the struggle is removed, so too are the conditions under which learning is built.

COGNITIVE TOIL

Cognitive toil refers to the effortful mental work involved in genuine learning: retrieving information from memory, resolving contradictions, generating explanations, and applying knowledge to unfamiliar problems. Bjork & Bjork (2020) term these *desirable difficulties* — conditions that slow down performance in the short term but produce more durable, transferable learning. They include retrieval practice (testing yourself), spaced repetition (distributing study over time), and interleaving (mixing different problem types). The essay argues that AI offloading eliminates precisely these conditions.

While AI companies search for toil to eliminate, we must ask whether learning can survive without it. Human intelligence still appears to require a metabolic price of admission. Toil is that price.

References

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